COOLING ENVIRONMENTS



Heat Reuse Subproject updates + Heat Reuse Economics tool

- General
- Reference Designs
- Policies / Outreach / Liaison
- Base Spec. Heat Reuse modules
- Heat Reuse Economics tool

Bharath Ramakrishnan *Microsoft*

Jaime Comella *DayOne*

Internal

OCP Heat Reuse Subproject essentials

- **Subproject's Leadership**: Bharath Ramakrishnan (Microsoft), Jaime Comella (DayOne), heatreuseleads@ocproject.net

- **Workstreams' Leadership:**
  - Policies: David Gardiner (David Gardiner and Associates), Petter Terenius (Uppsala University),
  - Economics: Jack Kolar (Dataquarium), Gemma Reeves (Alfa Laval)
  - Ref. Designs: Gabriel Lazar (Submer), Steven Schon (Quantacool)
  - Heat reuse module base spec working group:
    - Gemma Reeves (Alfa Laval), Mats Carselid (Danfoss)

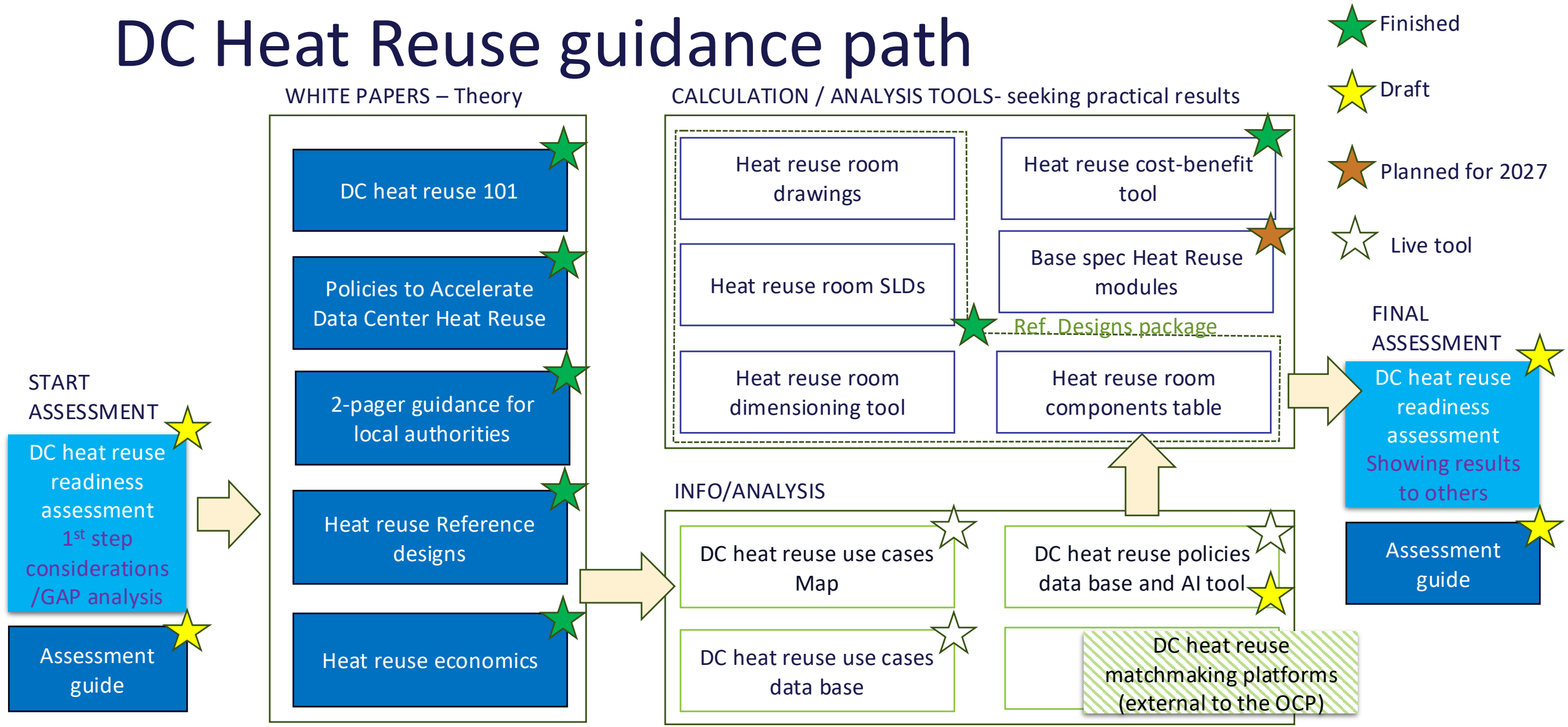
WIKI PAGE





- **Objectives**:
  - Gather experiences through guest presentations
  - Highlight challenges and opportunities, create awareness
  - Create guidance: whitepapers, practical tools, assessments.

DC Heat Reuse guidance path



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Reference Design Update

Gabriel Lazar *Submer*

Steven Schon *Quantacool*

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Reference Design Subproject

Current work is focused on two pillars

1. 24/7-type applications
    - Water treatment / desalination — white paper in progress
    - DAC being studied
2. Understanding the market
    - 1-hour interviews for projects 1 MW and above
        - Goal is to understand success factors and areas of focus
        - Output: conclusions for maturity mapping and next reference designs

Monthly meetings — first Thursday of every month. Please join!





1. Fresh Water Production & Facility Water Conservation



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Summary

Objective

- Find uses for data center low-grade waste heat with **year-round applicability**
  - Conventional un-boosted heat reuse has limited off-takers and seasonal demand

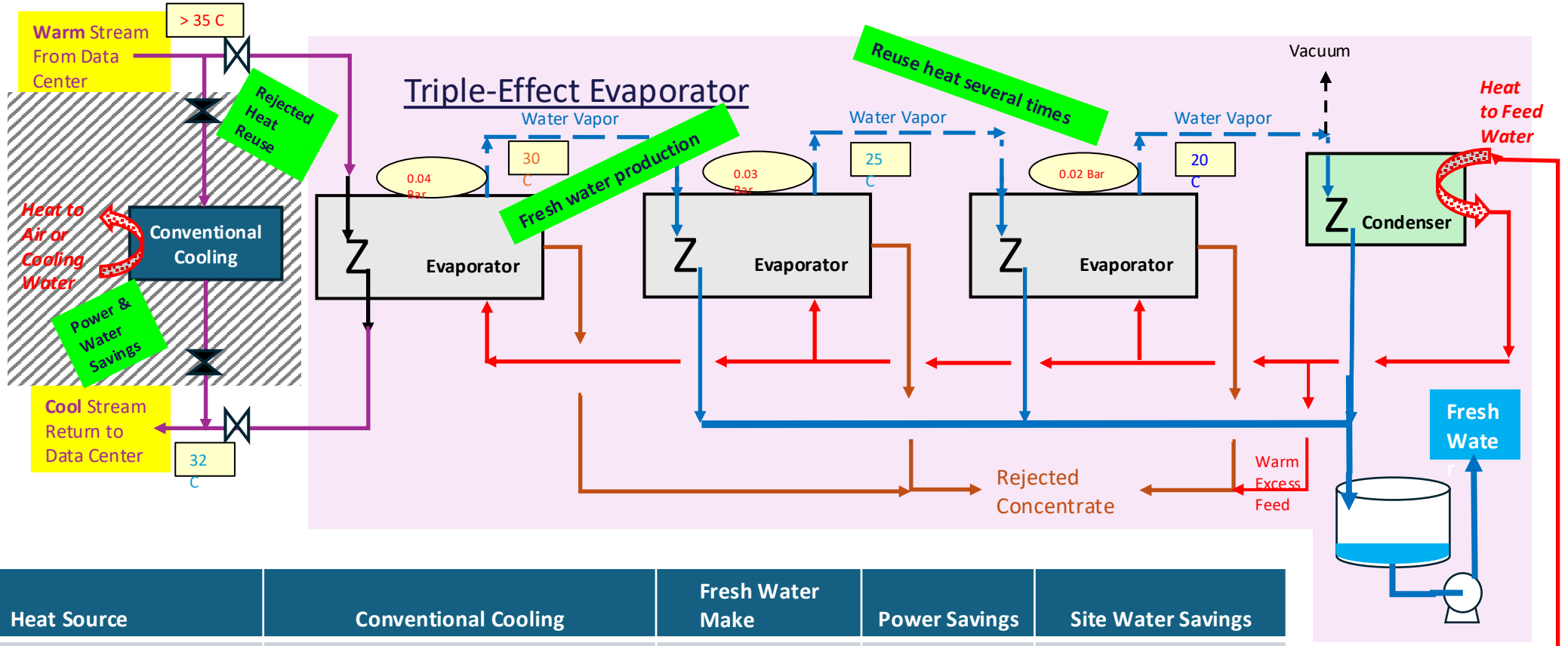
Proposition

- Study the use waste heat to **make fresh water from impure sources** (e.g. desalination)
- Reuse heat multiple times to increase water make (**Gain Ratio**)
  - Via multiple-effect evaporation (MEE)
  - Leverage higher working temperatures from liquid cooling
- Displace conventional data center cooling

Scope

- Direct TCS heat reuse instead of rejection to dry cooler
- Direct TCS heat reuse instead of rejection to facility water
- Indirect heat reuse from mechanical chiller (MEE instead of condenser)
  - *All three cases assume a liquid or 2-phase TCS cooling loop*

Diagram of heat reuse producing fresh water via MEE

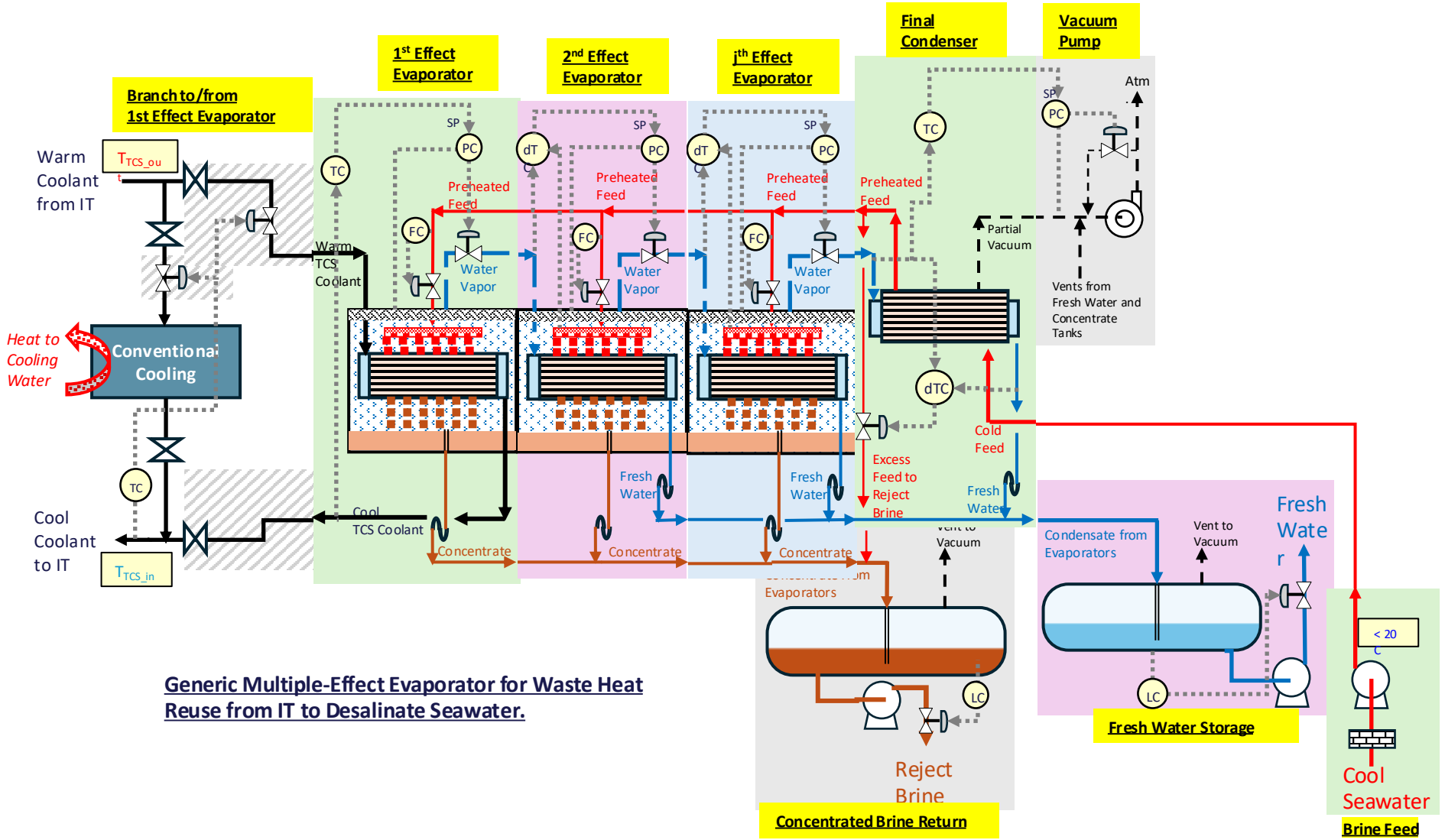


Heat Source	Conventional Cooling	Fresh Water Make	Power Savings	Site Water Savings
TCS (liquid or 2-phase)	Dry Cooler	X	X	
TCS (liquid or 2-phase)	FWS / Cooling Tower	X	X	X
TCS (liquid or 2-phase)	FWS / Dry Cooler	X	X	
Chiller (FWS or air)	HP refrigerant condenser	X		

Other Options Considered for Heat Reuse → Fresh Water

- Heat pump to increase ΔT → more stages, or legacy air-cooling heat
  - Incremental kW/m^3 water is greater than simply adding a reverse osmosis (RO) unit
- Preheat water (reduce viscosity) for conventional RO → reduce RO power use
  - < 4% energy recovery efficiency (kW RO power reduction / kW reused waste heat)
- Membrane distillation instead of MEE
  - Lower gain ratio than MEE (difficult to implement multiple effects)
  - More prone to fouling
  - Less mature technology; leader in MD exiting the business
- Use exhaust heat from onsite power generation to drive MEE
  - Could be very interesting and possibly more cost effective
  - Outside the scope of the OCP Heat Reuse subproject

MEE System Elements



Generic Multiple-Effect Evaporator for Waste Heat Reuse from IT to Desalinate Seawater.

Waste Heat Reuse to Produce Fresh Water via MEE

Benefits

- **Year-round demand** for heat reuse
  - Insensitive to data center variability
- **Fresh water from impure sources**
  - ~ 33 m^3/day per stage per MW of heat
  - Seawater, rivers, deep wells, etc.
- **Power savings**
  - Dry and wet cooler fans
  - Facility water pumps
  - Avoided power from external desalination
  - Potential carbon credits
- **Reduced / negative WUE**
  - Fresh water make; site water savings
- **Well-established technology**
  - Modularizable

Limitations / Concerns

- **Location constraints**
  - Proximity to a cool impure water source
  - Water-scarce region
  - Proximity to water utility / pipe network
- **Maximize DC return-water source ΔT**
  - Target 5 °C ΔT per stage
  - N stages = $(T_{TCS_cool} - T_{water_feed}) / 5$
- **Water off-taker collaboration**
  - Fresh water purity / treatment requirements
  - Service-level agreements
  - Value of fresh water ($/$m^3$)
- **Equipment complexity and cost**
- **Disposition / impact of concentrate**
- **New metrics needed?**
  - Net Water Usage Effectiveness
    - $NWUE = (W_{consumed} - W_{generated}) / E_{IT}$

Activities being carried out for desalination application

Status

- White paper v.1 in preparation
- Contacts and input from MEE system suppliers
- Process models for sizing equipment and utility requirements

Future Work

- Worked-out example (modular) designs
  - Direct heat reuse from TCS instead of rejection to dry cooler
  - Direct heat reuse from TCS instead of rejection to facility water
  - Indirect heat reuse from a mechanical chiller refrigeration condenser
- Cost estimation
- Tie-in to the heat reuse economics model

2. Industry Insights

Findings so far:



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What makes real heat reuse happen

Findings taken from: 5 MW university HPC site, its municipal district heating utility, and a large European colocation operator.

1 The operator leads

High engagement from the DC operator to bring the project to reality.

2 Higher temperatures

A 45 °C inlet removes need for chillers and lessens strain on heat pumps.

3 District heating decarb goals matter

Even if more heat is needed in winter, district heating networks move enough heat for year-round demand.

4 Broader stakeholder implication at local level

Municipalities, district heating, DC operator, universities got involved in advanced urban planning. Not a lone wolf strategy.

5 Champions are key

One project required five years of dedication from one person. And currently there are practically no economic incentives.

The pattern

The operator's major win is the brand equity gains in a time DCs are facing major pushback.

Learnings for future projects



IT cooling loop

Water loop is practically a requirement. Air gives up little heat.



Return temperature

45 °C inlet should make projects more feasible.



Ownership boundary

Need for collaboration and ownership on grey areas from the get go.



Heat pump

“Newish” technology that has not been proven for this application and uptime.



Off-taker

Size on recoverable heat, not facility load.

The gap the reference designs have to close: linking mature industries and techniques to heat reuse + showing the DC industry what is doable.

Call to Action

How to get involved

- Join the Heat Reuse subproject call: first Thursday of every month
- Contribute MEE supplier data, site constraints and water off-taker requirements, review v1 of whitepaper
- Second line of work looking into feasibility of carbon capture from BTH energy sources for those who are interested
- Bring your heat reuse projects to help drive the industry forward

Where to find additional information

- Heat Reuse wiki: **opencompute.org/wiki/Cooling_Environments/Heat_Reuse**
- Community page: **opencompute.org/community/heat-reuse**



HR Policies, Outreach and Liaison Update

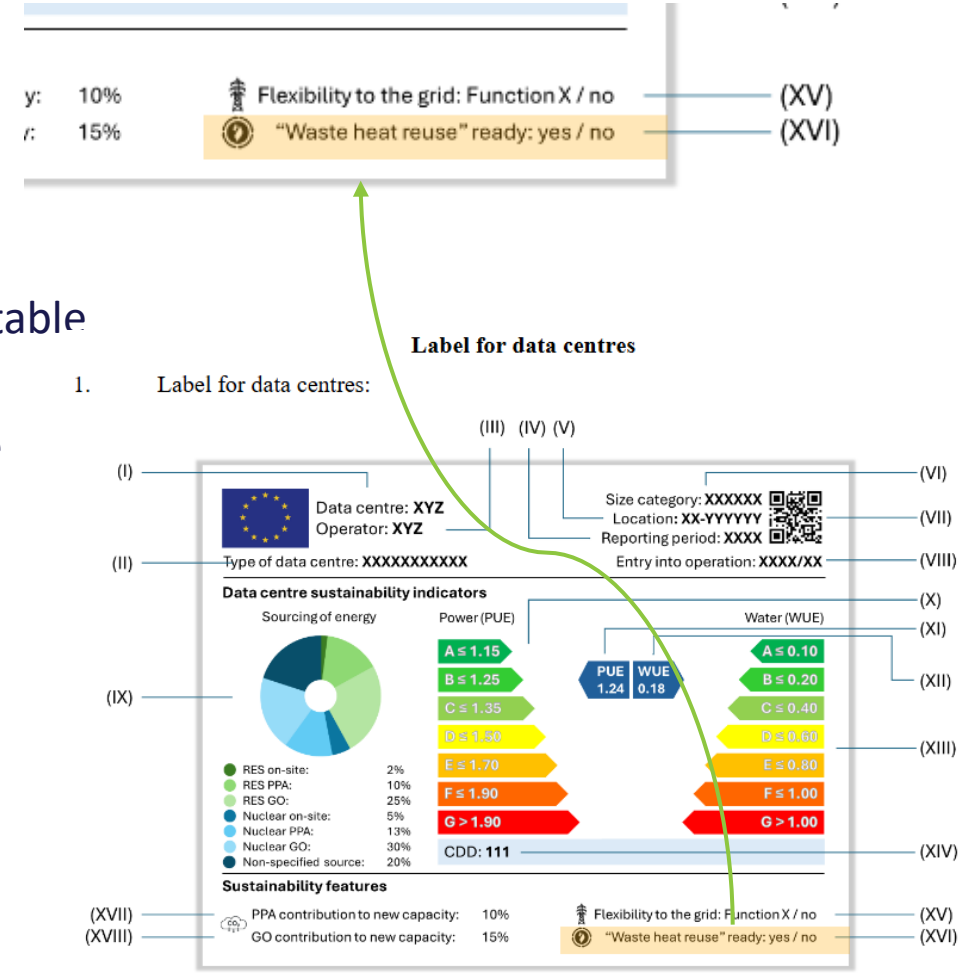
David Gardiner: CEO, David Gardiner and Associates, USA

Petter Terenius: Researcher at Uppsala University, Sweden

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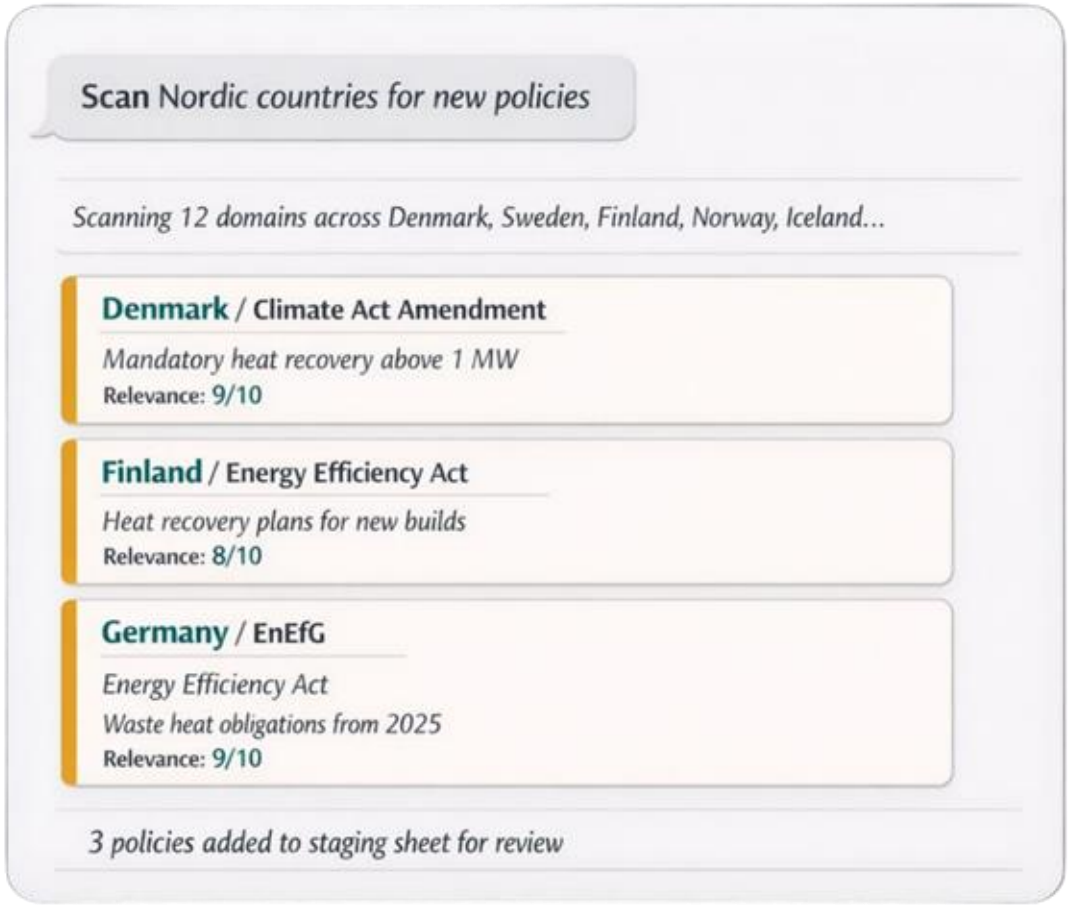
HR Policies, Outreach and Liaison Subproject

- Important policy updates
  - North America
    - Virginia passes first data center heat reuse legislation in U.S. Report due September 1 from its Department of Energy.
    - Pennsylvania Governor's office hosts industrial clean heat roundtable at which data center heat reuse is discussed as an option.
    - Significant policy maker concerns about electricity and water use leading to moratoriums or "pauses" on data centers.
  - Europe
    - EU Commission proposes to include data center heat reuse in its 2026 €1 billion Clean Industrial Heat Auction. Must use a heat pump and decarbonize industrial heat.
    - Heat reuse readiness mentioned in the EU data center label



HR Policies, Outreach and Liaison Subproject

- New AI tool
  - Searches for new heat reuse policies around the globe.
  - After human review, it helps to update the Working Group's Heat Reuse Policies Database.



HR Policies, Outreach and Liaison Subproject

- White Paper in progress
  - "Heat Reuse Ready“ Assessment for Data Centers
  - Self-assessment to evaluate how is one positioned to do heat reuse
  - Can be third-party certified for higher value
  - Old work discontinued 2 years ago
  - Now close to a requirement the EU and some US states

Checklist



White paper



Heat reuse readiness assessment-

Site and local assessment Profitability, heat demand matching, business model

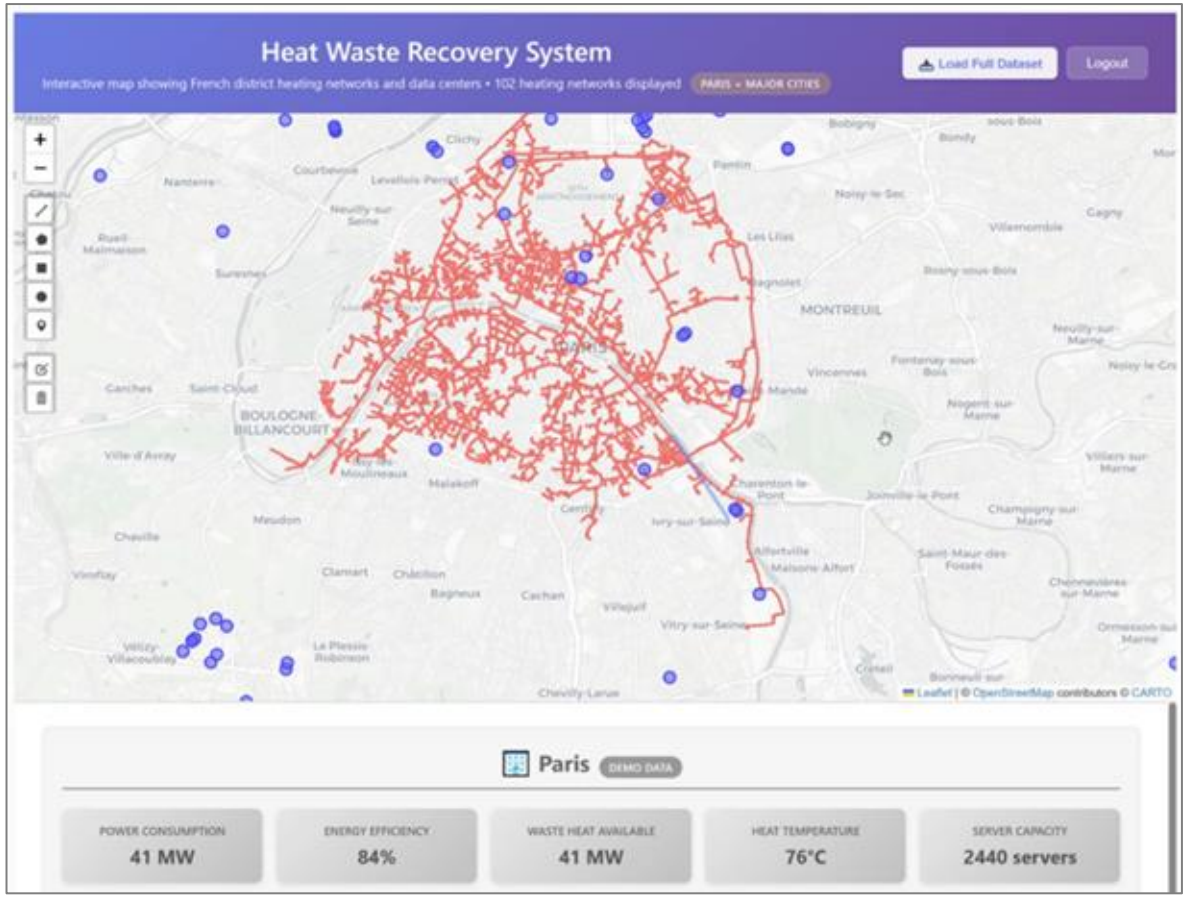
Attributes	Answer	Optimum parameter
SITE SELECTION		
A matchmaking platform has been used	Yes, No, N/A	Yes unless N/A
Location chosen based on real existing or planned heat reuse opportunities	Yes, No, N/A	Yes
Potential easements for the excess heat connection to the offtakers	Yes, No, N/A	No
REGIONAL ANALYSIS AND INCENTIVES		
Awareness of policies/regulations (local, regional, national, supranational)	Yes, No, N/A	Yes
Discussions with potential heat offtakers	Yes, No, N/A	Yes
Discussions with authorities (local, regional, national)	Yes, No	Yes
Community Due Diligence	Yes, No	Yes
HEATING DEMAND MATCHING		
Seasonality considered	Yes, No	Yes
Intraday variability considered	Yes, No	Yes
Short-term heat storage elements	12h or more, 8h, 4h or less, none	Any answer (between both attributes) different from "none"
Long-term heat storage elements	months, weeks, days, none	
Contracts with heat offtakers	0, 1, 2, 3, 4, more	the more the better
ECONOMIC VIABILITY		
Economic assessment done	Yes, No	Yes
Profitability ensured only with DC cooling savings	Yes, No	Yes
Profitability ensured in addition with heat sale	Yes, No	Yes
Profitability taking into account the "offtaker's" efforts	Yes, No	Yes
BUSINESS MODEL		
Maturity level	0, 1, 2, 3, 4, 5, 6	No optimum parameter
Quantifiable social benefit		

Heat reuse readiness assessment – Design

Attributes	Answer	Optimum parameter
SPACE AND DESIGN OF THE INTERFACES		
Dedicated room/space for heat recovery	Yes, No	Yes
Preliminary design of the alocated room/space	Yes, No	Yes
Size of the room/reserved space	≥ 56 m2 per MW OR ≤ 56 m2 per MW for ≤ 1 MW excess heat ≥ 37 m2 per MW OR ≤ 37 m2 per MW for ≥ 1 MW excess heat	≥ 56 m2 per MW for ≤ 1 MW excess heat ≥ 37 m2 per MW for ≥ 1 MW excess heat
Allocation and availability for ready tie-in to power, network, cooling connections	Yes, No	Yes
Space reserved for other potential machines (heat pumps)	<10 m2 per MW; ≥ 10 m2 per MW	≥ 10 m2 per MW
Space reserved for other potential auxiliary systems (storage tanks)	<80 m3 per MWh; ≥ 80 m3 per MWh	≥ 80 m3 per MWh
Natural temperature level of the provided excess heat (prior to boosting)	≤ 30°C, between 30°C and 40°C, between 40°C and 50°C, between 50°C and 60°C, ≥ 60°C	≥ 60°C
Physical state of the excess heat that can be provided	Water; water + glycol; liquid (specify in the notes column); air; condensable vapor; gas (specify in the notes column); hybrid (specify)	water
Own usage of excess heat	Yes, No	Yes
Operation of data center depends on heat extraction by third parties or vice-versa	Yes, No	No
Standards and Norms alignment		
Modularity of the design	Yes, No	Yes

HR Policies, Outreach and Liaison Subproject

- The Heat Reuse Platform project
  - Objective: Match data centers with nearby offtakers in Europe
  - Joint work with European Data Centre Association, EuroHeat & Power, Net Zero Innovation Hub, large data center owners
  - Matchmaking tool in beta stage. Please contact Petter Terenius (OCP project representative) if interested in DC business in Europe.
  - https://hrp.eudca.org/



Call to Action

- Heat Reuse Platform project
  - https://www.linkedin.com/company/the-heat-reuse-platform/about
  - Please share your performance data with the platform!



- OCP HR Subprojects:
  - Homepage: https://www.opencompute.org/projects/heat-reuse



  - Details: https://www.opencompute.org/w/index.php?title=Cooling_Environments/Heat_Reuse
  - Please join our discussions!



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Heat Reuse Module Base Specification

Gemma Reeves: Alfa Laval

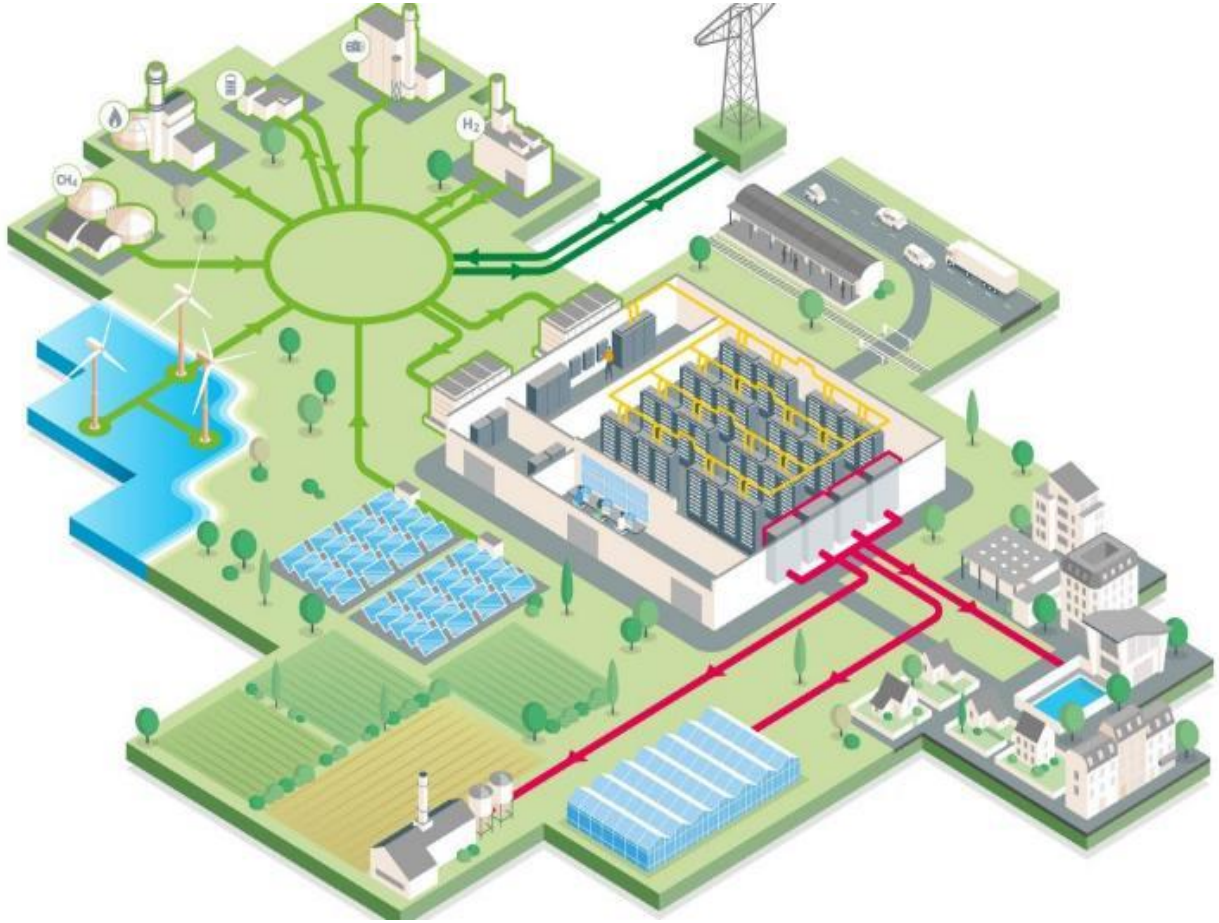
Mats Carselid: Danfoss

OCP SILVER Member

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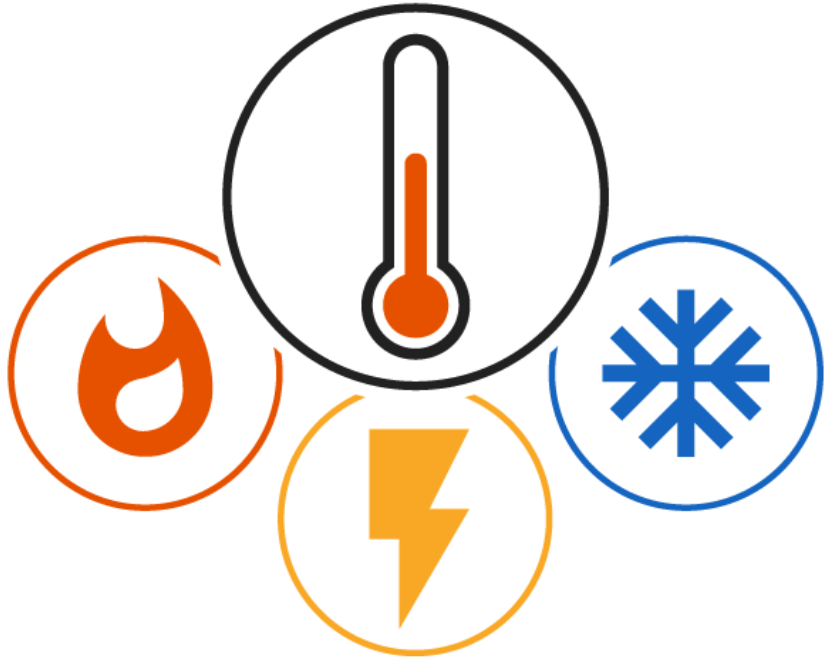
Introduction

- Purpose of Base Specification
- What is a Heat Reuse Module (HRM)?
- How it Works
- Typical Heat Reuse Applications



A HRM ensures

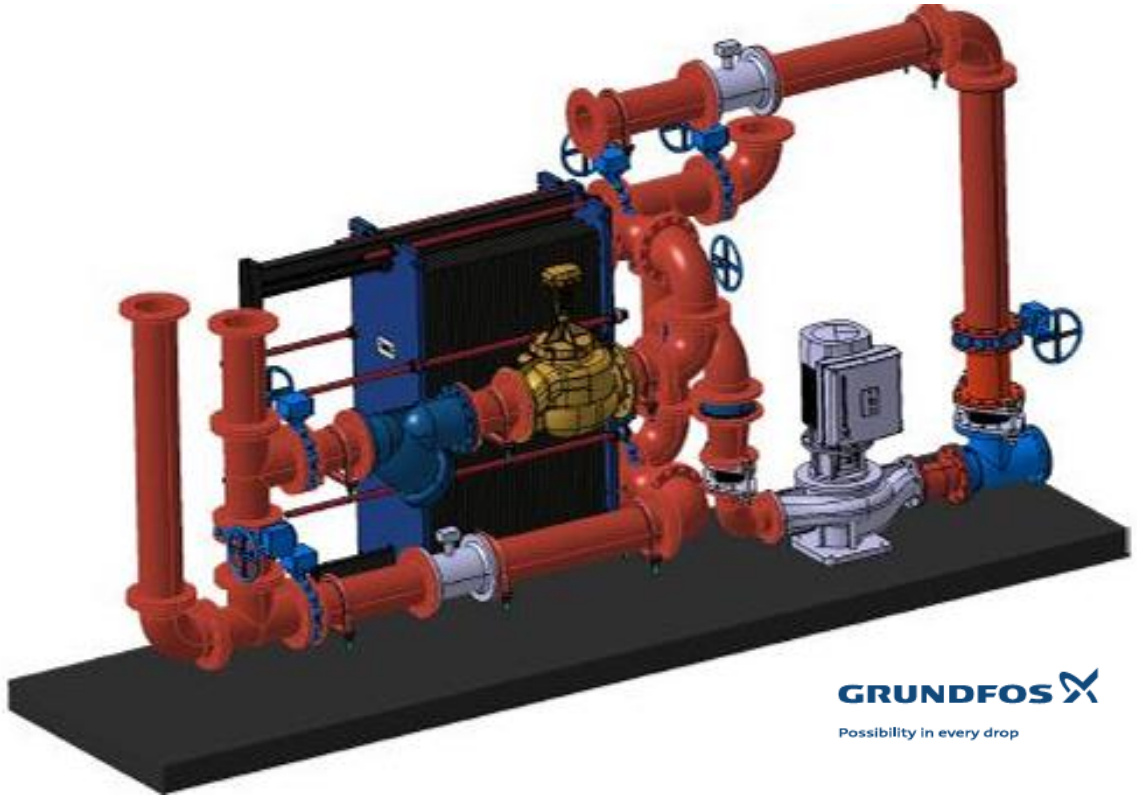
- **Hydraulic separation** between data-centre loops and heat-reuse loops.
- **Efficient transfer of thermal energy** at defined ΔT values.
- **Safe pressure and temperature operation** across all circuits.
- **Interoperability** with existing facility water loops and external customers.



This base specification is focused on Passive heat-reuse modules

What is in an HRM?

- A Plate heat exchanger
- Pumps and hydraulic assemblies.
- Filtration and water-quality control.
- Instrumentation, metering and automated valves.
- Controls
- A module frame/skid.



Heat Reuse Base Specification Current Sections

- **Environmental / Regulatory Requirements**
  - Pressure Vessel / Heat Pump / District-Heating Interface Requirements
  - Quality Standards
  - Environmental & Safety Standards
- **Module Components and Subsystems**
  - Heat Exchanger
  - Pumping and Hydraulics
  - Filtration
  - instrumentation & Controls
  - Skid/Frame Requirements
  - Insulation Requirements
- **Design & Performance Requirements**
  - Operating Conditions
  - System Performance
  - Interface Requirements – Data Centre Loop
  - Interface Requirements – Heat Reuse Loop
  - Water Quality Requirements
- **Testing, Commissioning & Documentation**
  - Factory Acceptance Testing
  - Site Acceptance Testing
  - Documentation Requirements

Work Group Approach and Call to Action

- Small focused dynamic break out groups/meetings
  - Pumps
  - Controls and Metering
  - Piping
  - Heat Exchangers
  - Fittings and Valves
- Defined targets for progress milestones
- Target to be accepted/presented at OCP in 2027
- Contact to get involved
  - Gemma Reeves @ Alfa Laval : Gemma.Reeves@AlfaLaval.com
  - Mats Carselid @ Danfoss: Mats.Carselid@Danfoss.com













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Heat Reuse Economics tool

Jack Kolar: Dataquarium

Gemma Reeves: Alfa Laval

Internal

Heat Reuse Economics Tool

WHAT IT IS:

- This tool gives instant responses to enquiries regarding the values of equipment selected to deliver a liquid stream to harvest heat from a data center. It could, for example point out differences in performance between a system with a 2 degree approach to that of a 5 degree approach
- This tool has been created in a way to be accessible to users at many levels:
    - "Beginner" (non-coding),
    - "Professional" (makes business-level changes), or
    - "Developer" (can edit code when necessary)

Heat Reuse Economics Tool

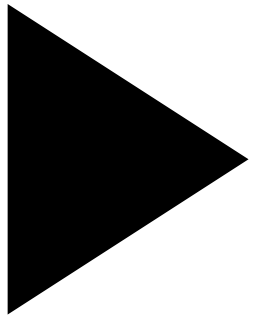
WHY THIS MATTERS:

- The tool can be used for business purposes, or can be contributed to by, anyone with knowledge of the industry.

- No "coding" knowledge is required for simple enquiries
- When the costing data is called into question, it can be easily adjusted
- Should refinements be desired to the calculation methods, they are completely accessible

Heat Reuse Economics Tool

LIVE DEMO



Call to Action

- Watch the demo.

- Run your own Demo:
- https://github.com/opencomputeproject/OCP-CE-HR-Economics-Tool



- The tool is in an immature stage - waiting for further contributions
- It will remain in Github as a "Living" document
- As the industry begins to grow, data will be added as it is learned.

Heat Reuse Subproject Call to action

- **Present your use case // Join our workstreams // Suggest new workstreams // Use our tools // Leave comments**
- Contribute to the edition of the map: Is your data center spotted? // Is your data center not spotted yet?
- Latest guidance:

  Fresh water production with heat reuse // Heat reuse readiness checklist;
  AI search tool for policies // Heat reuse modules base spec // Economics tool

WIKI PAGE

Heat reuse
Zoom calls link!

Calls (always 5PM CET, 11AM ET, 8AM PT)

1st Tuesday of the month: HR Policies call
1st Thursday of the month: HR Reference Designs call
2nd Thursday of the month: HR Economics call (7AM PT!)
3rd Tuesday of the month: General HR call

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Thank You!



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